

Science Olympiad — ScioVirtual Day 1 Homework

Names of participants: (Please print neatly)

School Name: _____ Team Number: _____

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Scoring: Time to solve first problem: _____ (use to calculate Bonus below)

Question	Value	Incorrect letters	Deduction	Score
Timed	250			
1	245		≡	
2	267		≡	
3	277		≡	
4	294		≡	
Timed Bonus	Timed:			
Final Score				

Timed Question [250 points] The message below was encoded into the DNA sequence of a synthetic organism. It translates to a quote, encrypted using an Aristocrat Cipher. What does it say? When you have solved it, raise your hand so that the time can be recorded and the solution checked.

NOWS ZVEU NHIS OB ZVE SIFOUW VL NAS RVEULSZ NAHB

RETZ. NAS GUEHNB VG ZVEU TOFVU QHTT BVVL FS BAVQL.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	5			6	3	2	4	2			4		6	4		2	2	7	4	6	10	2			5
Replacement																										

1) **[245 points]** Your lab partner discovered this pattern on the back of a graphing calculator. It's a quote, encoded as an Aristocrat using a K1 alphabet. What was the K1 Key Phrase used to encode it?

Enter the Key Phrase here:

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AZWH REBJ F RZL PSXJDBT, F RSXUV LYBJV MEB BJMFTB

LWESSU VZP RZMWEFJD MEBLB IBJ TXJ FJ WFTWUBL ZTSXJV

MEB YFMWE.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	11		2	7	7		1	1	9		5	6			2		4	5	5	3	4	6	4	2	5
Replacement																										

2) **[267 points]** Found inside a dissected frog's stomach (plastic, don't worry): this Aristocrat note using a K1 alphabet. What was the K1 Keyword used to encode it?

Enter the Keyword here:

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O RQZDK XH KHKD LYS QZZW MA NZ NUK DWOKD. LJZEK XK

CKDND L FUZQK FZCQS FLNOYT MA PZC XK NZ POYLQQH

RZXX UZXX.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	2		3	6	1	3		3		1	12	5	2	5	5	2	6	2	2	1	3		2	5	3	11
Replacement																										

Enter the Keyword here:

WSIH NLIHI. AWE'HI REFQA NLCN M'U JIIRMVK VMFI NWGCA.

[illegible]

Enter the Keyword here:

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IZW MDENV?

[illegible]